

MINGRUI LI

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EDUCATION

EMORY UNIVERSITY, Rollins School of Public Health

Master of Public Health, Biostatistics

Courses: Applied Linear Models, Statistical Inference, Machine Learning, Data Science Toolkit, SAS & R Programming

Atlanta, GA
08/2024-05/2026

LIVERPOOL UNIVERSITY

Bachelor of Science, Bioinformatics

Courses: Genomics & Proteomics, Computational Biology, Molecular Genetics, Database Systems, Probability Theory

Liverpool, UK
08/2020-05/2024

SKILLS

- **Programming:** SAS, R, Python, Linux, SQL, STATA, MATLAB
- **Tools:** Tableau, Power BI, Docker, Git Jupyter Notebook, Microsoft Office Suite, AWS, Azure
- **Methods:** Regression, Survival Analysis, PCA, t-SNE, Random Forest, XGBoost, LSTM, Model Evaluation

PROFESSIONAL EXPERIENCE

Emory University

Research Assistant

Atlanta, GA
01/2024-Present

- Designed and maintained an **interactive analytics dashboard** using **R Shiny** to explore high-dimensional datasets across dozens of categories, enabling non-technical users to perform self-service data exploration and insight generation
- Built **end-to-end data pipelines** in **R** and **SQL** to extract, clean, and integrate large-scale structured data from a **cloud-hosted relational database**, supporting scalable analysis across thousands of observations and tens of thousands of features
- Applied **multivariable linear regression models** to quantify relationships between key predictors and outcomes, incorporating covariate adjustment and **multiple testing correction** to ensure statistically valid inference
- Automated **data preprocessing and analysis workflows**, including normalization, feature filtering, batch-effect correction, and model result generation, producing **reproducible analytical outputs** used across multiple projects

NBG-Virogin Biotech

Bioinformatics Analyst - Internship

Shanghai, China
06/2022-09/2022

- Processed and standardized clinical and biomarker datasets from over 10,000 oncology patients to support biomarker validation for treatment response prediction
- Engineered a **Random Forest** model to assess associations between biomarkers and therapy outcomes, tuning hyperparameters via cross-validation to achieve over 90% **AUC** and identify high-risk subgroups for targeted treatment
- Applied **PCA** and variable clustering to reduce over 300 lab and molecular features into interpretable dimensions, facilitating discussions with clinical researchers on variable importance and feature redundancy
- Maintained and updated an existing analysis workflow in **Python** and **R** for a biomarker validation project, incorporating new patient data, re-running predictive models, and generating updated visual reports and case-level summaries for weekly clinical team meetings

Xi'an Jiaotong Liverpool University

Genomics Bioinformatics Analyst

Suzhou, China
08/2023-05/2024

- Analyzed 196 urban metagenomic samples to profile microbial composition using **Kraken2** and reduced dimensionality with **PCA** and **t-SNE** to identify regional clustering patterns
- Built an **XGBoost** model to predict resistance risk from microbial and land-use features, achieving 89% accuracy and identifying Bacteroides and Pseudomonas as key predictors for intervention planning
- Applied **LSTM time series modeling** to forecast resistance gene prevalence under varying pesticide exposure scenarios, supporting early warning efforts in environmental policy planning
- Mapped 15+ antibiotic resistance genes concentrated in high-pesticide zones, generating data-driven recommendations for environmental health policy discussions

RESEARCH EXPERIENCE

Emory University

EEG Biomarker Validation in Epilepsy Drug Trials

Atlanta, GA
01/2025-11/2025

- Cleaned and integrated multi-center epilepsy trial datasets by correcting input errors, standardizing date formats, and constructing patient-level timelines for seizure event tracking
- Performed survival analysis using **SAS** procedures to compare time-to-first-seizure across treatment groups, and evaluated **EEG** spike activity as a predictive biomarker by calculating sensitivity, specificity, positive predictive value, and negative predictive value
- Designed logic-based classification algorithms to categorize **EEG** results and assessed inter-rater reliability among physicians using **Cohen's Kappa statistic**
- Delivered structured analytical datasets and summary tables to demonstrate both drug efficacy and the diagnostic utility of EEG monitoring in early seizure detection

PUBLICATIONS

Chen, M.†, Li, M.†, Li, R. Y., Jiang, J., Qin, Z. S. (2026).

Age Effect Explorer: A Shiny application to browse and visualize tissue-specific age-related gene expression changes.

Bioinformatics Advances, vbag026. <https://doi.org/10.1093/bioadv/vbag026>

(Co-first author; equal contribution)